

Problem-Solution Fit

Date	15 March 2026
Team ID	xxxxxxx
Project Name	Credit Card Approval Prediction Using Machine Learning
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

1. CUSTOMER SEGMENT(S) [CS] • Banks and financial institutions • Credit card applicants • Bank credit officers • Loan/credit approval departments	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] • Limited knowledge of approval criteria • Time constraints • Incomplete applicant information • Budget limitations for advanced software • Dependence on manual verification	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Rule-Based Approval Systems Pros • Fast processing • Easy to implement Cons • Cannot learn from data • Limited accuracy • Rigid decision rules
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] • Manual review of applications is slow (High) • Human errors in approval decisions (Medium)	9. PROBLEM ROOT / CAUSE [RC] • Manual evaluation process • Large number of applications	7. BEHAVIOR + ITS INTENSITY [BE] • Frequently submits applications online (High) • Checks application status repeatedly (High)

- Inconsistent approval decisions (**High**)
- Long waiting time for applicants (**High**)
- Difficulty assessing large volumes of applications (**High**)

- Complex financial eligibility criteria
- Human decision-making inconsistencies
- ☐ Time-consuming document verification

- Expects quick approval decisions (**High**)
- Provides financial documents (**Medium**)
- ☐ Contacts customer support if delayed (**Medium**)

3. TRIGGERS TO ACT [TR]

- Increasing number of credit card applications
- Need for faster customer service
- Rising operational costs
- Demand for accurate risk assessment
- Digital transformation in banking

4. EMOTIONS BEFORE / AFTER [EM]

Before

- Frustrated
- Anxious
- Uncertain
- Stressed

After

- Confident
- Satisfied
- Relieved
- Trustworthy

10. YOUR SOLUTION [SL]

Develop a **Machine Learning-based Credit Card Approval Prediction System** that automatically analyzes applicant information and predicts approval decisions. The solution reduces manual effort, improves accuracy and consistency, shortens processing time, and provides a user-friendly web application for efficient credit card approval management.

8. CHANNELS OF BEHAVIOR [CH]

ONLINE

- Bank Website
- Mobile Banking App
- Email Notifications
- SMS Alerts

OFFLINE

- Bank Branch
- Customer Service Desk
- Phone Support